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Synthesis of hyperbranched polyethers with diverse structures based on "$A_1 + B_2$" ring-opening polymerization

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ABSTRACT

The research on hyperbranched polyethers (HBPEths) currently focused on two main areas: (1) the development of a practical synthesis method and (2) the functionalization of HBPEths. "$A_1 + B_2$" ring-opening polymerization ("$A_1 + B_2$"-ROP) offers an efficient approach to access functional HBPEths with controllable structures. In the present study, a series of HBPEths were successfully prepared by employing "$A_1 + B_2$"-ROP, where the A_1 and B_2 represent a monofunctional primary alcohol and a biepoxy monomer, respectively. The obtained HBPEths possessed molar masses ranging from 2600 to 30000 g/mol. By employing functional A_1 primary alcohol initiators, it is possible to directly introduce a single type or a mixture of functionalities including fluoride, alkyne, alkene, azide, and furfuryl groups into the HBPEths terminal structures, while by adjusting various B_2 monomers with different linkage between both epoxy groups the main chain structures could be well controlled. Moreover, apart from conventional end group modification via esterification, the functionalities can be introduced during the one-pot "$A_1 + B_2$"-ROP process via ether bonds, which are stable against hydrolysis and redox-reaction.

1. Introduction

The hyperbranched polymers (HBPs) belong to a class of highly branched macromolecules possessing three-dimensional structures and highly irregular terminal groups [1–5]. In contrast to linear structures, the HBPs exhibit enhanced solubility and fluidity attributed to mild entanglement between their chains, resulting from their three-dimensional structure. Over the past decade, HBPs have become a focal point of extensive interdisciplinary research. Hyperbranched polyethers (HBPEths) incorporating polyethylene oxide (PEO) segments either in the main or side chains have advantages of low glass transition temperature, exceptional solubility, low viscosity, and good biocompatibility [6–11]. These distinctive characteristics render HBPEths highly versatile for a wide range of potential applications in various fields, such as drug delivery, plastic modification, adhesives, cross-linking agents, and coatings [12–16].

Based on the distinct reaction mechanism, the synthesis of HBPEths can be categorized into two strategies: (1) self-condensing ring-opening polymerization (SCROP) of inimers containing both epoxy and pendant hydroxyl groups [17–21], and (2) "$A_2 + B_3$" ring-opening polymerization ("$A_2 + B_3$"-ROP) employing biepoxy compound as the A_2 monomer and triol as the B_3 monomer [22–26]. In 1999, Frey and coworkers [17] have developed SCROP by employing partially deprotonated 1,1,1-*tris*-(hydroxymethyl)propane as an initiator and glycidol as an inimer. Under slow addition conditions, HBPEths with molar masses in the range of 1250–6500 g/mol, M_w/M_n of 1.13–1.47, and degree of branching (DB) values of 0.53–0.59 were successfully obtained. Simultaneously, Fréchet et al. [12] first reported the "$A_2 + B_3$"-ROP of 1,2,7,8-diepoxyoctane as the A_2 monomer and 1,1,1-tris(hydroxymethyl)ethane as the B_3 monomer for accessing HBPEths. Jia et al [27]. employed the same strategy to synthesize a novel main-chain thermo-responsive HBPEth, that exhibited a variable lower critical solution temperature (LCST), ranging from 19.0 to 40.3 °C by adjusting the hydrophilic/hydrophobic balance of 1,4-butanediol diglycidyl ether and triol initiator.

The key properties of HBPEth, such as solubility, polarity, reactivity, viscosity, and thermal properties, are significantly influenced by their terminal groups. The majority of end groups in HBPEths obtained through both the abovementioned strategies are hydroxyl groups, which

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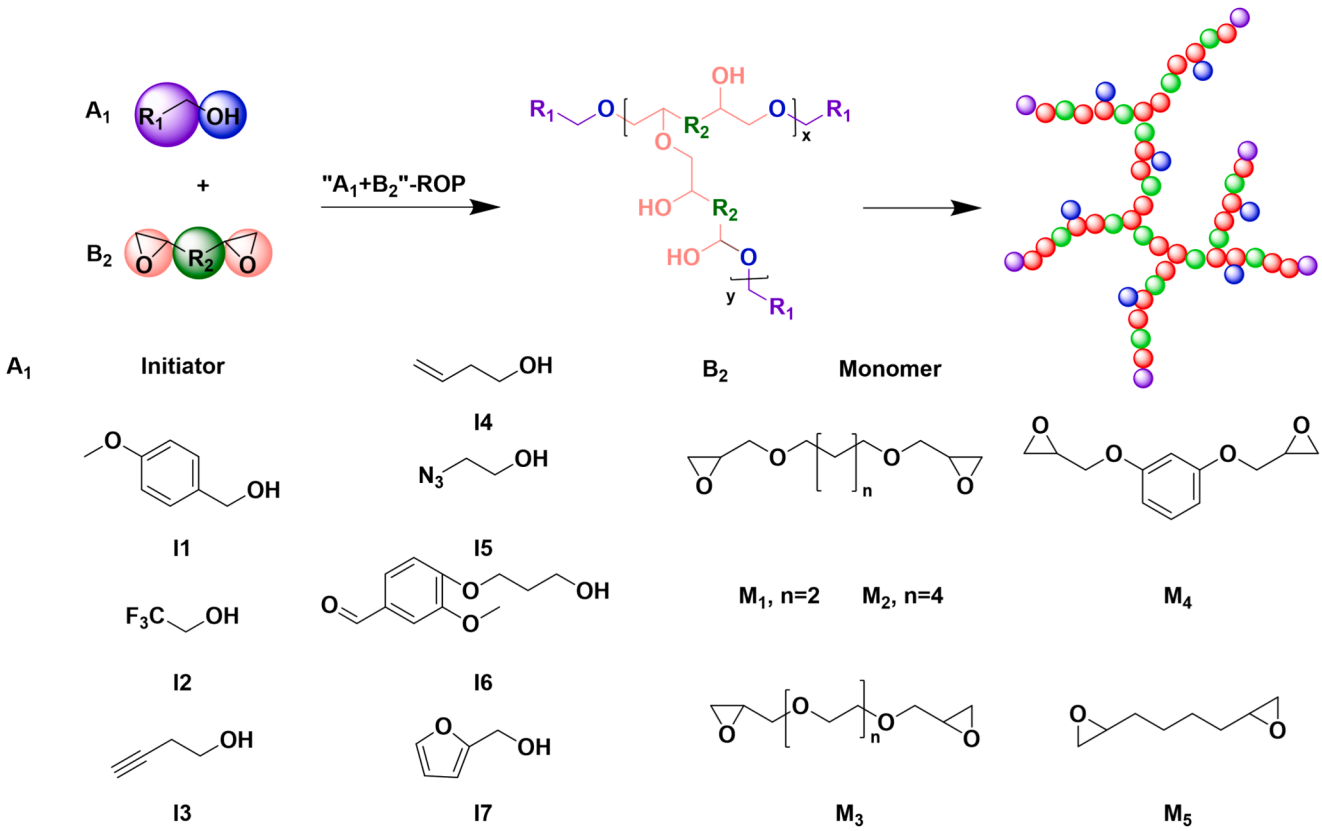


Scheme 1. Synthetic route of HBPEths with varying main chain and terminal group structures.

can be modified via esterification reactions with carbonyl compounds such as anhydride, carboxylic acid, or carbonyl chloride to connect functional groups with the HBPEth main chain [14,28–30]. Xiang et al. [31] produced a water-soluble HBPEth photosensitizer by esterifying HBPEth in a single step. Jana and coworkers [32] introduced gallate, tyrosine, and trehalose functionalities into HBEPths via esterification of hydroxyl terminal groups and investigated their potential in inhibiting lysozyme/huntingtin protein aggregation under intra-/extracellular space. However, the modified end groups bonded via ester linkage are generally unstable, and can undergo hydrolysis under acidic or basic conditions. Etherification serves as an alternative strategy for end group modification of HBPEths. Koschek et al. [33] successfully achieved methylation with controlled degrees on the hydroxyl groups of HBPEth through a phase-transfer catalysis approach. However, this method necessitated the use of a large amount (1.0 eq. per OH group) of highly toxic dimethyl sulfate, and entailed a total reaction/purification time exceeding one week. From the point of functional HBPEth development, there is an urgent need to develop a simple method for preparing HBPEth with controlled structures and functionalities.

In our previous study, "$A_1 + B_2$" ring-opening polymerization ("$A_1 + B_2$"-ROP) has been successfully developed, where A_1 represents a monofunctional primary alcohol initiator and B_2 denotes a bicyclic monomer, respectively [34]. A series of hyperbranched polycarbonates with precisely controlled end group compositions and high molar masses was synthesized by employing one-pot organo-catalyzed "$A_1 + B_2$"-ROP under mild polymerization conditions. In the present work, we report on the anionic "$A_1 + B_2$"-ROP of primary alcohol as the A_1 initiator with biepoxy compound as the B_2 monomer, targeting HBPEths, as shown in Scheme 1. The structure diversity of the HBPEths can be tailored by choosing the linkage within the biepoxy monomer to determine the main chain structure or using a functional alcohol initiator to introduce functional end groups, such as alkenyl, alkyne, azide, and furfuryl functionalities. Moreover, apart from conventional end group modification via esterification, the functionalities can be introduced during the one-pot "$A_1 + B_2$"-ROP process via ether bonds, which are stable against hydrolysis and redox-reaction.

2. Experimental section

2.1. Materials

2-Azido-1-ethanol (96 %+, Adamas), 1,4-butanediol diglycidyl ether (99 %, Aladdin), 3-butyn-1-ol (98 %, Energy Chemical), 3-buten-1-ol (98 %, Macklin), 1,2,7,8-diepoxyoctane (97 %, Macklin), diglycidyl resorcinol ether (94 %, Heowns), dimethyl sulfoxide (DMSO, 99.8 %, Extra Dry, with molecular sieves, Water $\leq$ 50 ppm (by K.F.), Energy Chemical), 1,6-hexanediol diglycidyl ether (99 %, Macklin), poly (ethylene glycol) diglycidyl ether (99 %, Macklin), furfuryl alcohol (99 %, Adamas), sodium hydride (60 %, Adamas), 2,2,2-trifluoroethyl alcohol (98 %, Aladdin), tetrahydrofuran (99.5 %, Energy Chemical) were commercially available and used as received. 4-Methoxybenzyl alcohol (MeOBnOH, 98 %, Aladdin) was purified by recrystallization from *n*-hexane/chloroform (v/v = 1/5).

2.2. Measurements

^{1}H and ^{13}C nuclear magnetic resonance (NMR) spectra were recorded using a BRUKER AVANCE 500 MHz at 500 and 125 MHz, respectively. Chloroform-*d* ($CDCl_3$-*d*, 99.8 D%, Energy Chemical) and *d*imethylsulfoxide-d_6 (DMSO-d_6, 99.5 D%, Energy Chemical) were used as solvents for NMR measurements. Fourier transform infrared (FTIR) spectra were collected on a BRUKER VERTEX70 infrared instrument with an attenuated total reflectance (ATR) attachment from 4000 to 600 cm^{-1}. The molar masses and dispersities ($Đ_M$) were analyzed employing an EcoSEC HLC-8320 size exclusion chromatography (SEC) at 40 °C. The system was operated at a flow rate of 0.35 mL/min with THF as the

solvent. Calibration was performed using polystyrene (PS) standards. Differential scanning calorimetry (DSC) was performed with a Netzsch DSC 204 F1 Phoenix at a heating/cooling rate of 10 K min^{-1} under a nitrogen atmosphere.

2.3. General procedure for the synthesis of HBPEths by "$A_1 + B_2$"-ROP

The obtained HBPEths are designated as HBPEth-Mx-Ix, where Mx and Ix indicate the used diglycidyl ether monomer (M1-M5) and primary alcohol initiator (I1-I7), for example the HBPEth prepared from 1,4-butanediol diglycidyl ether (M1) initiated by 4-methoxybenzyl alcohol (I1) is named as HBPEth-M1-I1. In the representative preparation of HBPEth-M1-I1, in a 10 mL Schlenk tube, sodium hydride (40 mg, 60 % purity dispersed in mineral oil, corresponding to 1.00 mmol of active NaH) was suspended in DMSO (1 mL). After adding 4-methoxybenzyl alcohol (I1, 0.13 mL, 1.00 mmol) under an argon atmosphere, the mixture was stirred at room temperature for 2.5 h to finish the deprotonation. M1 (0.35 mL, 1.00 mmol) was added into the mixture at 40 °C under an argon atmosphere. The polymerization was quenched by the addition of a mixture of water (0.5 mL)/ethanol (2.5 mL). The crude product was purified by dialysis using 1000 Da membrane to give a translucent, pale yellow material.

^{1}H NMR (400 MHz, Dimethyl Sulfoxide-*d*): δ(ppm) = 7.24 (**CH_{Ar}**-C-CH_2), 6.88 (**CH_{Ar}**-C-O), 4.78–4.54 (CH-**OH**), 4.38 (Ar-**CH_2**-O), 3.73 (O-**CH_3**), 3.68–3.20 (**CH_2**-O and **CH**-O), 1.48 (O-CH_2-**CH_2**). IR (ATR, cm^{-1}): 3440 (O-H), 2923 and 2866 (C-H), 1610 and 1514 (C=C_{Ar}), 1247 (C_{Ar}-O), 1095 (C-O).

HBPEth-M1-I2. HBPEth-M1-I2 was synthesized in a similar manner to HBPEth-M1-I1 but with 2,2,2-trifluoromethyl alcohol (I2) instead of I1.

^{1}H NMR (400 MHz, Dimethyl Sulfoxide-d_6): δ(ppm) = 4.95–4.65 (CH-**OH**), 4.12–4.00 (CF_3-**CH_2**-O), 3.75–3.23 (**CH_2**-O and **CH**-O), 1.51 (O-CH_2-**CH_2**). IR (ATR, cm^{-1}): 3419 (O-H), 2929 and 2871 (C-H), 1419 (C-F), 1114 (C-O).

HBPEth-M1-I3. HBPEth-M1-I3 was synthesized in a similar manner to HBPEth-M1-I1 but with 3-butyn-1-ol (I3) instead of I1.

^{1}H NMR (400 MHz, Dimethyl Sulfoxide-d_6): δ(ppm) = 4.88–4.42 (CH-**OH**), 4.08–3.60 (**CH**≡C-CH_2-**CH_2**-O) 3.60–3.22 (**CH_2**-O and **CH**-O), 1.82 (CH≡C-**CH_2**), 1.52 (O-CH_2-**CH_2**). IR (ATR, cm^{-1}): 3415 (O-H), 2923 and 2824 (C-H), 2293 (C≡C), 1093 (C-O).

HBPEth-M1-I4. HBPEth-M1-I4 was synthesized in a similar manner to HBPEth-M1-I1 but with 3-buten-1-ol (I4) instead of I1.

^{1}H NMR (400 MHz, Dimethyl Sulfoxide-d_6): δ(ppm) = 5.85–5.72 (CH_2=**CH**), 5.11–4.94 (**CH_2**=CH), 4.73–4.53 (CH-**OH**), 3.72–3.59 (O-**CH_2**-CH_2-CH=CH_2), 3.59–3.22 (**CH_2**-O and **CH**-O), 2.25 (CH_2=CH-**CH_2**), 1.51 (O-CH_2-**CH_2**). IR (ATR, cm^{-1}): 3450 (O-H), 2923 and 2864 (C-H), 1641 (C=C), 1109 (C-O).

HBPEth-M1-I5. HBPEth-M1-I5 was synthesized in a similar manner to HBPEth-M1-I1 but with 2-azido-1-ethanol (I5) instead of I1.

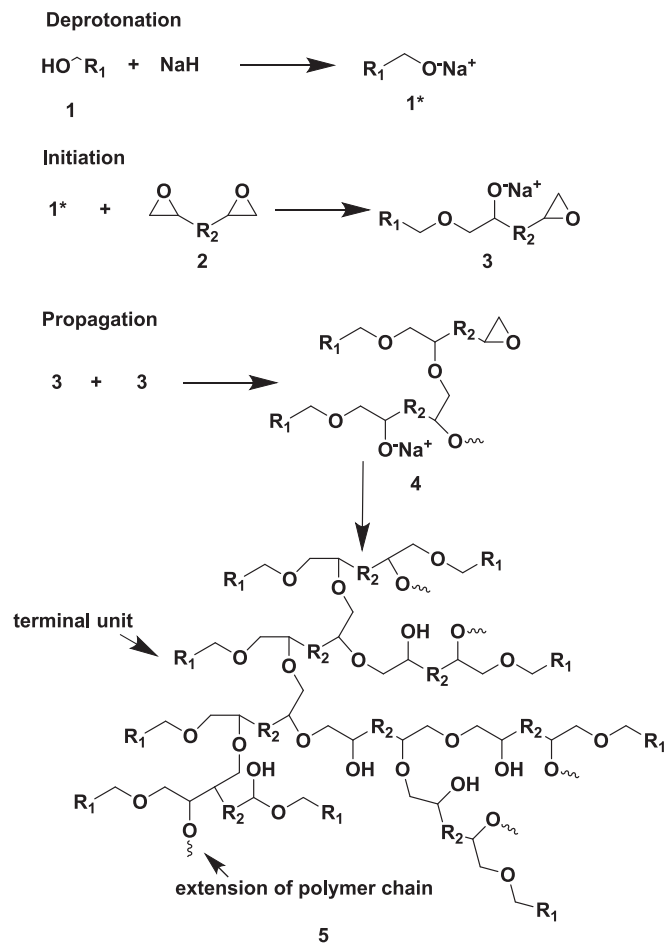
^{1}H NMR (400 MHz, Dimethyl Sulfoxide-d_6): δ(ppm) = 4.88–4.62 (CH-**OH**), 3.74–3.62 (N_3-CH_2-**CH_2**-O), 3.62–3.22 (N_3-**CH_2**, **CH_2**-O and **CH**-O) 1.52 (O-CH_2-**CH_2**). IR (ATR, cm^{-1}): 3415 (O-H), 2927 and 2868 (C-H), 2102 (N=N=N), 1662 and 1598 (C-N), 1109 (C-O).

HBPEth-M1-I6. HBPEth-M1-I6 was synthesized in a similar manner to HBPEth-M1-I1 but with modified vanillin (I6) instead of I1.

^{1}H NMR (400 MHz, Dimethyl Sulfoxide-d_6): δ(ppm) = 7.39–6.75 (**CH_{Ar}**), 4.83–4.48 (CH-**OH**), 4.38 (Ar-O-**CH_2**), 4.07–3.92 (Ar-O-CH_2-CH_2-**CH_2**-O), 3.82–3.70 (**CH_2**-O), 3.59–3.22 (**CH_2**-O and **CH**-O), 1.99–1.88 (Ar-O-CH_2-**CH_2**), 1.49 (O-CH_2-**CH_2**). IR (ATR, cm^{-1}): 3396 (O-H), 2923 and 2860 (C-H), 1600 and 1514 (C=C_{Ar}), 1263 (C_{Ar}-O), 1105 (C-O).

HBPEth-M1-I7. HBPEth-M1-I7 was synthesized in a similar manner to HBPEth-M1-I1 but with furfuryl alcohol (I7) instead of I1.

^{1}H NMR (400 MHz, Dimethyl Sulfoxide-d_6): δ(ppm) = 7.62 (CH=**CH**-O), 6.40 (**CH-CH**), 4.81–4.56 (CH-**OH**), 4.40 (ox-**CH_2**-O), 3.72–3.58 (ox-CH_2-O-**CH_2**), 3.58–3.22 (**CH_2**-O and **CH**-O), 1.49 (O-CH_2-**CH_2**). IR (ATR, cm^{-1}): 3456 (O-H), 2920 and 2862 (C-H), 2360 and 2336 (C=C), 1263 (C_{Ar}-O), 1087 (C-O).



Scheme 2. Possible mechanism of the "$A_1 + B_2$"-ROP of biepoxy monomers.

HBPEth-M1-Imix (I2 + I3 + I7**).** HBPEth-M1-Imix was synthesized in a similar manner to HBPEth-M1-I1 but with a mixture composed of I2 (40.36 μL, 0.4 mmol), I3 (35.47 μL, 0.4 mmol) and I7 (34.88 μL, 0.4 mmol) instead of I1.

^{1}H NMR (400 MHz, Dimethyl Sulfoxide-d_6): δ(ppm) = 7.62 (CH=**CH**-O), 6.40 (**CH-CH**), 5.86–5.72 (CH_2=**CH**), 5.10–4.95 (**CH_2**=CH), 4.81–4.56 (CH-**OH**), 4.40 (ox-**CH_2**-O), 4.11–3.99 (**CH_2**-CF_3), 3.76–3.22 (**CH_2**-O and **CH**-O), 2.24 (CH_2=CH-**CH_2**), 1.51 (O-CH_2-**CH_2**). IR (ATR, cm^{-1}): 3435 (O-H), 2923 and 2866, (C-H), 2360 and 2336 (C=C_{ox}), 1458 (C=C), 1276 (C-F), 1105 (C-O).

HBPEth-M2-I3. HBPEth-M2-I3 was synthesized in a similar manner to HBPEthM1-I3 but with M2 instead of M1.

^{1}H NMR (400 MHz, Dimethyl Sulfoxide-*d*): δ(ppm) = 4.60–4.20 (CH-**OH**), 4.08–3.22 (**CH**≡C-CH_2-**CH_2**-O, **CH_2**-O and **CH**-O), 1.82 (CH≡C-**CH_2**), 1.47 (O-CH_2-**CH_2**), 1.29 (CH_2-**CH_2**-**CH_2**-CH_2). IR (ATR, cm^{-1}): 3465 (O-H), 2927 and 2860 (C-H), 2354 and 2243 (C≡C), 1097 (C-O).

HBPEth-M3-I3. HBPEth-M3-I3 was synthesized in a similar manner to HBPEth-M1-I3 but with M3 instead of M1.

^{1}H NMR (400 MHz, Dimethyl Sulfoxide-*d*): δ(ppm) = 4.75–4.30 (CH-**OH**), 4.08–3.28 (**CH**≡C-CH_2-**CH_2**-O, **CH_2**-O and **CH**-O), 1.82 (CH≡C-**CH_2**). IR (ATR, cm^{-1}): 3514 (O-H), 2871 (C-H), 1955 (C≡C), 1103 (C-O).

HBPEth-M4-I3. HBPEth-M4-I3 was synthesized in a similar manner to HBPEth-M1-I3 but with M4 instead of M1.

^{1}H NMR (400 MHz, Dimethyl Sulfoxide-*d*): δ(ppm) = 7.12 (**CH_{Ar}**), 6.48 (O-C-**CH_{Ar}**), 5.00–4.50 (CH-**OH**), 4.15–3.40 (**CH**≡C-CH_2-**CH_2**-O, **CH_2**-O and **CH**-O), 1.80 (CH≡C-**CH_2**). IR (ATR, cm^{-1}): 3446 (O-H), 2923

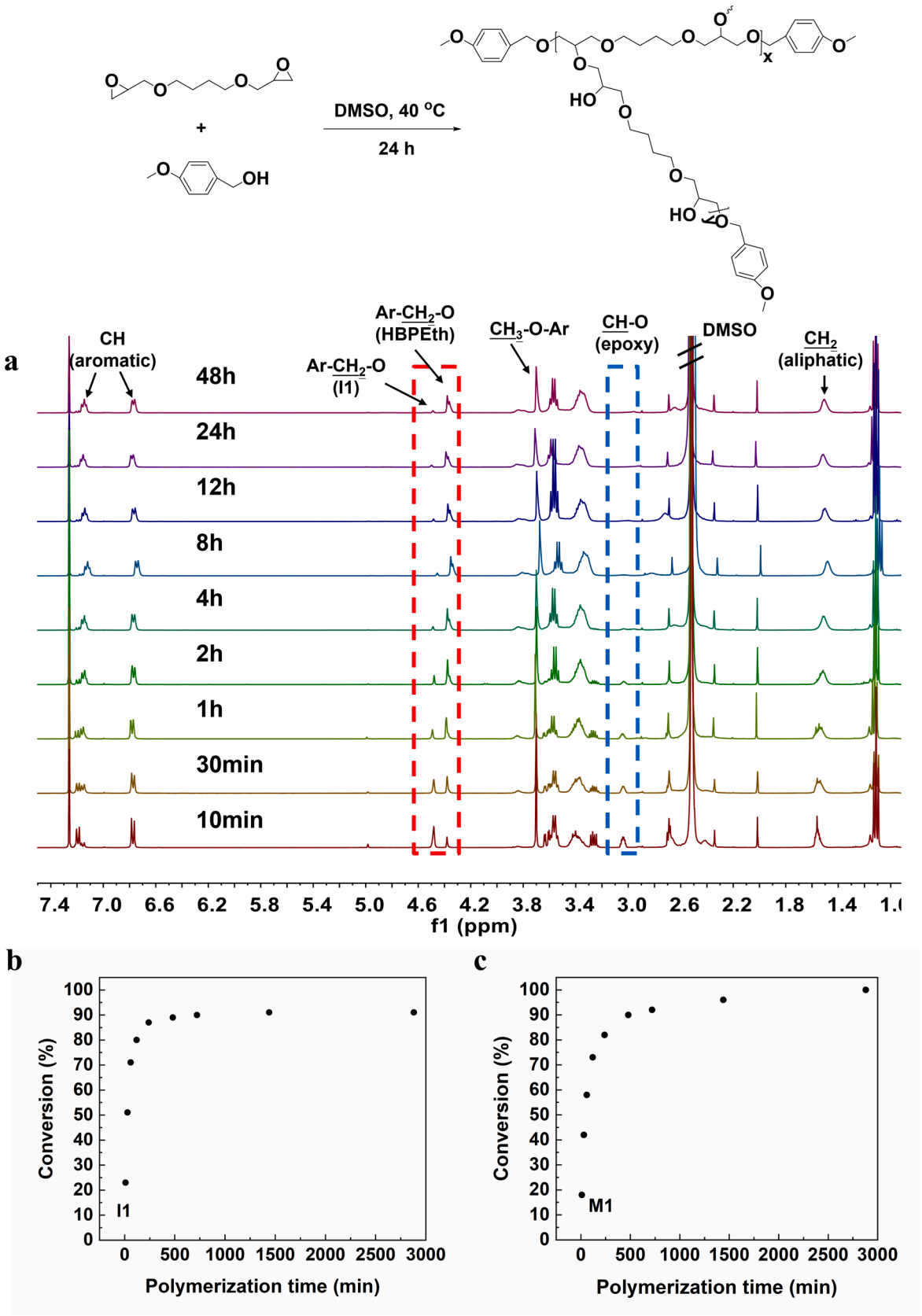


Fig. 1. (a) ^{1}H NMR spectra of the reaction solution of HBPEth-M1-I1 at different polymerization times; conversions of I1 (b) and the epoxy groups (c) during the "A_1 + B_2"-ROP.

and 2860 (C-H), 2229 (C≡C), 1598 and 1456 (C=C_{Ar}), 1147 (C_{Ar}-O), 1085 (C-O).

HBPEth-M5-I3. HBPEth-M5-I3 was synthesized in a similar manner to HBPEth-M1-I3 but with M5 instead of M1.

^{1}H NMR (400 MHz, Dimethyl Sulfoxide-*d*): δ(ppm) = 4.07–3.15 (CH≡C-CH_2-**CH_2**, **CH_2**-O and **CH**-O), 1.82 (CH≡C-**CH_2**), 1.42 (CH_2-**CH_2**-**CH_2**-CH_2), 1.24 (CH-**CH_2**). IR (ATR, cm^{-1}): 3406 (O-H), 2929 and 2862 (C-H), 2341 and 2237 (C≡C), 1091 (C-O).

3. Results and discussion

3.1. Results of synthesis of HBPEths using 1,4-butanediol diglycidyl ether (M1)

The "$A_1 + B_2$"-ROP of biepoxy monomers, initiated by deprotonated primary alcohols according to the proposed mechanism, is illustrated in Scheme 2. The primary alcohol (**1**) is first deprotonated by sodium hydride to generate the initiator **1***, which can subsequently react with the biepoxy monomer and open one of its epoxide rings resulting in the formation of a secondary alkoxide intermediate **3**. Ideally, propagation can occur similar to a SCROP mechanism by nucleophilic attack of the secondary alkoxide on an intermolecular epoxide of **3** [35] to give a HBPEth structure. According to the "$A_1 + B_2$"-ROP mechanism, it is crucial to control the feed ratio between initiator and biepoxy monomer at approximately 1: 1 to achieve higher molar mass and prevent undesired cross-linking side reactions.

The conditions of "$A_1 + B_2$"-ROP were optimized by using 1,4-butanediol diglycidyl ether (M1) as biepoxy monomer initiated by deprotonated 4-methoxybenzyl alcohol (MeOBnOH). The selection of solvents for this "$A_1 + B_2$"-ROP was based on several critical requirements: (1) high solubility for the initiator (alkoxide), monomer, and resulting hyperbranched polyether; (2) high melting point to prevent solvent evaporation during the polymerization process; (3) aprotic nature to avoid interference with the anionic polymerization mechanism; (4) chemical inertness in the presence of alkoxide species. Considering the sustainability and toxicity, DMSO was chosen as the polymerization solvent. Satisfying the premise of high conversion of epoxy groups, the polymerization should be carried out under lower concentration and temperature conditions to reduce the polymerization rate, enhance the controllability and suppress the cross-linking side reaction [34]. Hence, the "$A_1 + B_2$"-ROP was conducted in a DMSO solution with a concentration of 1.0 M at 40 °C. Different monomer-to-initiator ratios were evaluated and the results are shown in Table S1. The optimal feed ratio of M1: MeOBnOH was determined to be 1: 1.2. When the ratio was less than 1: 1.2 (HBPEth-M1-I1-4 and 5), undesired cross-linking side reactions were observed, while further enhancing the feed ratios to 1:1.3 and 1:1.4 led to continuous decrease of molar masses. Furthermore, slow addition of M1 using a syringe pump was also investigated with the molar feed ratio of [M1]: [I1] = 1: 1.2. The resulting HBPEth possessed a M_n of 1200 g/mol and a slightly increased dispersity of 2.5. These results indicate that the slow addition technique provided minimal advantages compared to the "one-pot" process. To gain insights into the polymerization process, the kinetics of "$A_1 + B_2$"-ROP with M1: MeOBnOH = 1.2: 1 were investigated by employing ^{1}H NMR spectroscopy (Fig. 1). Samples after specified polymerization time was taken from the polymerization mixture and terminated using water/ethanol. As depicted in Fig. 1a, as the polymerization proceeded, the characteristic signal corresponding to Ar-**CH_2**-O^- at 4.48 ppm decreased, whereas the HBPEth end group Ar-**CH_2**-O signal at 4.38 ppm increased, providing direct evidence of the reaction between I1 and M1. In addition, due to epoxide opening, there was a continuous decrease in the characteristic CH signal at 3.04 ppm for epoxide groups. Conversion (Conv.) of I1 was calculated using equation (1) by comparing the relative peak intensity of the Ar-**CH_2**-O end group ($I_{4.38\ ppm}$) with that of total Ar-**CH_2**-O of MeOBnOH ($I_{4.30-4.60\ ppm}$). Similarly, conversion of epoxy groups in M1 was obtained by comparing the unreacted CH signal ($I_{3.04\ ppm}$) with both inert aliphatic CH_2 groups ($I_{1.56\ ppm}$).

Table 1
Results of the synthesis of HBPEth-M1s.[a]

Sample	[M1]: [I][b]	DB_{Frey}[c]	DB_{Suzuki}[c]	End group ratio [I]: [OH]	M_w (g/mol)[d]	$Đ_M$[d]
HBPEth-M1-I1	1.0: 1.2	0.61	0.44	50: 50	3000	2.31
HBPEth-M1-I2	1.0: 1.0	0.60	0.43	51: 49	4100	1.47
HBPEth-M1-I3	1.0: 1.1	0.57	0.40	49: 51	6000	1.86
HBPEth-M1-I4	1.0: 1.1	0.66	0.49	45: 55	29,300	5.98
HBPEth-M1-I5	1.0: 1.0	0.59	0.42	51: 49	2600	1.28
HBPEth-M1-I6[e]	1.0: 1.0	0.39	0.32	35: 65	1600	1.09
HBPEth-M1-I7	1.0: 1.0	0.66	0.50	47: 53	5200	3.09
HBPEth-M1-I_{mix}	1.0: 1.2	0.66	0.50	47: 53	6100	3.07

[a] Conditions: polymerization was conducted at 40 °C for 24 h.
[b] Initiator was deprotonated by using NaH for 2.5 h in DMSO before adding M1.
[c] Calculated by ^{1}H NMR spectroscopy in DMSO-d_6.
[d] Determined by SEC in THF.
[e] Aldehyde end group was not observed in the ^{1}H NMR spectrum.

$$\text{Conv.(I1)} = \frac{I_{4.38ppm}}{I_{4.30-4.60ppm}} \times 100\% \quad (1)$$

$$\text{Conv.(epoxy groups)} = 1 - \frac{I_{3.04ppm}}{I_{1.56ppm}/2} \times 100\% \quad (2)$$

As illustrated in Fig. 1b, I1 was rapidly consumed within the initial 2 h of polymerization, achieving a conversion of 80 %. After 24 h, the conversion of I1 reached its maximum value of 91 %. M1 exhibited similar behavior throughout the polymerization process. The conversions of epoxy groups were observed to be 19.6 %, 57.3 %, and 97.2 % after 10 min, 1 h, and 12 h, respectively (Fig. 1c). Notably, the conversion of epoxy groups continued to increase even after 12 h and reached nearly 100 % after 24 h. In contrast, no change in I1 conversion was observed, indicating that the reaction between secondary alkoxide and epoxide dominated the system, which is consistent with the proposed "$A_1 + B_2$"-ROP mechanism. Kinetic analysis revealed that the reactions of I1 and M1 follow second-order kinetics, with rate constants determined as $k_{I1} = 0.02669$ ($R^2 = 0.98309$) and $k_{M1} = 0.01875$ ($R^2 = 0.98711$), respectively. Moreover, during the first 12 h of polymerization, there was a delay in epoxide conversion compared to that of I1. This could be attributed to a small amount of water inevitably introduced into the polymerization mixture while weighing NaH and adding M1 and DMSO. Therefore, an excess of initiator is necessary for rapid generation of secondary alkoxide and elimination of undesired water.

HBPEths with functional terminal groups are attractive in the field of polymer chemistry to obtain specialty materials or for further post-polymerization modification through click-chemistry, such as alkyne-azide reaction, thiol-ene reaction, Diels-Alder reaction etc., enabling precise conjugation of desired functionalities, for instance, binding bioactive ligands/drugs or stimuli-responsive moieties to achieve pH-, redox-, or enzyme-triggered drug release under mild conditions while maintaining biocompatibility [30,36,37]. To achieve this goal, a variety of functional primary alcoholic initiators containing fluoro-alkyl (I2), alkynyl (I3), alkenyl (I4), azide (I5), aldehyde (I6), and furfuryl (I7) functional groups was used in the "$A_1 + B_2$"-ROP for the straightforward preparation of functional HBPEths. As illustrated in Table 1, the [M1]: [I] ratio was varied from 1.0: 1.0 to 1.0: 1.2 to avoid cross-linking reaction due to the different reactivity of initiators. HBPEths with molar

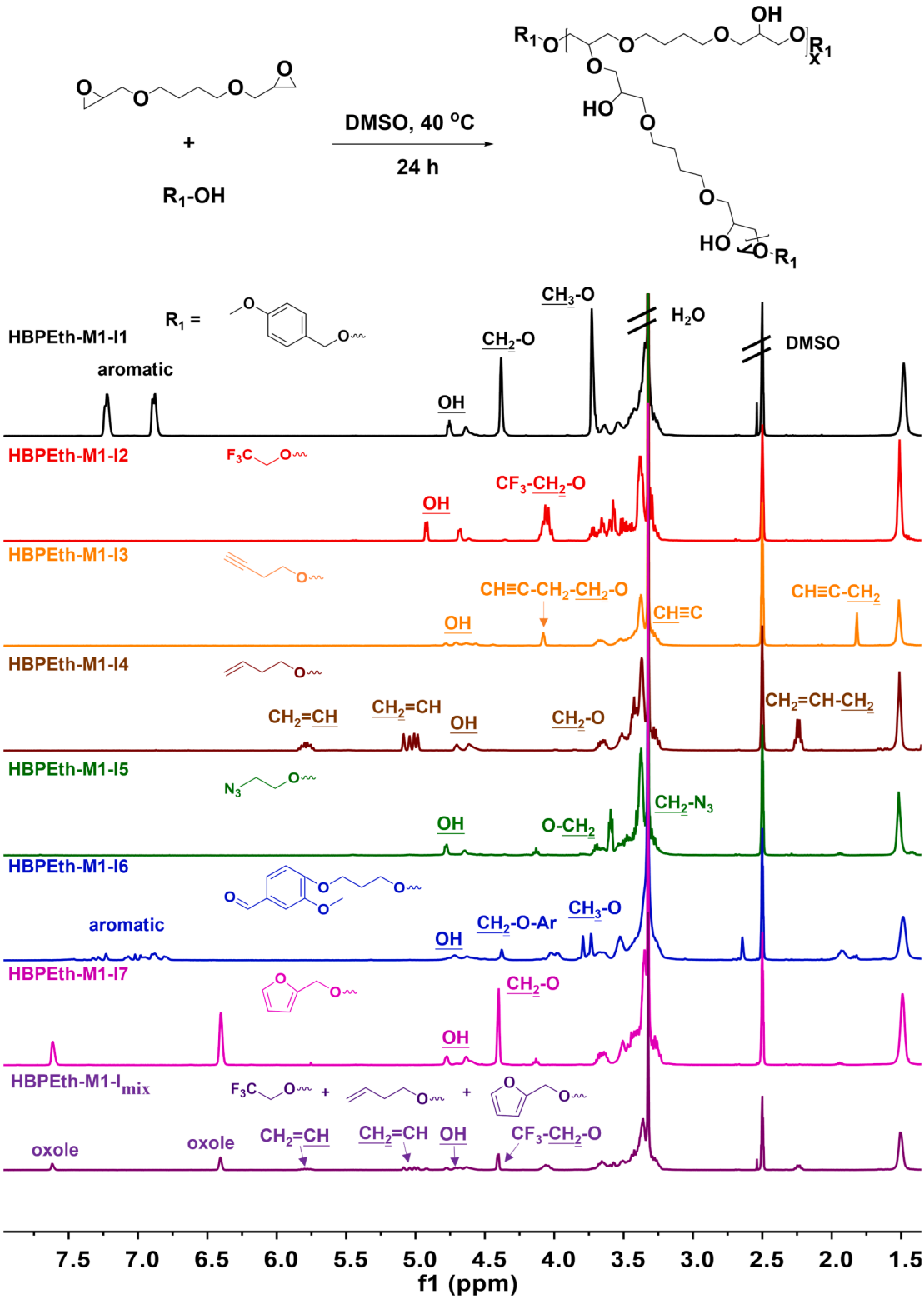


Fig. 2. ^{1}H NMR spectra of HBPEth-M1s initiated by different functional primary alcohol.

masses (M_w) in the range of 2600–29300 g/mol were successfully obtained. Since HBPs with abundant hydroxyl end groups often exhibit aggregation resulting in smaller dynamic radius and strong interaction with the SEC columns [22,33], the measured molar masses generally underestimate their actual values. The results of ^{1}H NMR (Fig. 2) and IR (Figs. S1-S8) spectroscopies were consistent with the expected chemical structures of HBPEths. The signals in the range of 3.22–3.68 ppm were corresponded to the polyether (CH_2-O) backbone, and those of aliphatic methylene (CH_2-CH_2) groups appeared at 1.45–1.55 ppm. The functional end groups could be clearly identified in the ^{1}H NMR spectra: 6.83–7.22 ppm (aromatic, HBPEth-M1-I1), 4.00–4.12 ppm ($\underline{CH_2}$-CF_3, HBPEth-M1-I2); 1.78–1.84 ppm ($\underline{CH_2}$-CH≡C, HBPEth-M1-I3), 4.94–5.11 ppm ($\underline{CH_2}$=CH, HBPEth-M1-I4), 3.62–3.74 (N_3-CH_2-$\underline{CH_2}$-O, HBPEth-M1-I5), 6.75–7.39 (aromatic, HBPEth-M1-I6) and 7.60 (oxole,

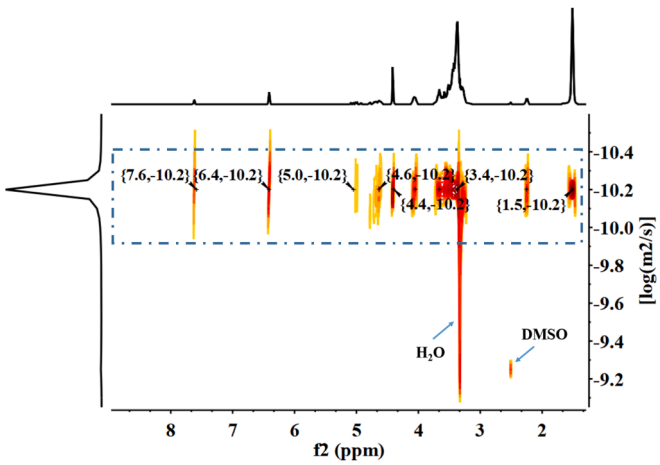


Fig. 3. DOSY spectrum of HBPEth-M1-Imix.

HBPEth-M1-I7). Notably, the presence of the aldehyde end group in HBPEth-M1-I6 was not observed in the ^{1}H NMR spectrum probably due to a side reaction between the aldehyde and alokoxide or NaH. This suggests that caution should be exercised when selecting an initiator, considering potential side reactions under strong alkaline environment. Besides this, all functional groups were successfully incorporated into the HBPEth backbone via stable ether bonds. No signal around 3.00 ppm was observed in all cases indicating a high conversion of B_2 monomers and the absence of epoxy end group. The end group ratios of [I]: [OH] were determined to be close to 50 % as expected, by comparing the relative intensities between initiator and hydroxyl end groups in ^{1}H NMR spectra in DMSO-d_6. To obtain HBPEth terminated with multiple functionalities, a mixture of I2, I4, and I7 was employed in the "$A_1 + B_2$"-ROP of M1. In the ^{1}H NMR spectrum of HBPEth-M1-I_{mix}, distinct characteristic signals corresponding to each functionality can be clearly observed. The DOSY spectrum (Fig. 3) reveals that all terminal group signals exhibit identical diffusion coefficient, suggesting the successful incorporation of multiple functionalities into the same HBPEth backbone.

The degree of branching (DB) measures the suitability of a polymerization method for constructing dendritic structures. Taking HBPEth-M1-I1 as an example for the structure analysis, as shown in Scheme 3, a HBPEth structure is composed of dendritic units (D), linear units (L) with hydroxyl side groups, and two possible terminal units (T) with initiator end groups (T_1) and unreacted epoxy end groups (T_2). According to the NMR analysis, the T_2 unit can be neglected owing to the high conversion of epoxy groups. DB values can be calculated based on the relative intensities of signals corresponding to respective subunits in ^{1}H NMR spectra using the following equation (Eq. 3) derived by Frey et al. [38].

$$DB_{Frey} = \frac{2D}{2D + L} \tag{3}$$

The signals at 4.60–4.90 ppm were assigned to the secondary hydroxyl side groups corresponding to the linear units, while the peak at 1.50 ppm represented the total of dendritic and linear units in the branching structure. As each biepoxy monomer contains two branching structures, the contents of D, L, and T units can be calculated as follows: D = I(1.50 ppm)/4–I(4.60–4.90)/2, L = I(4.60–4.90)/2 and T = T_1/2 = I (initiator end group signal)/2. The DB_{Frey} values for HBPEth-M1 samples ranged between 0.59 and 0.66, which is close to the theoretical value of 0.66 [39] for HBPEth preparation via AB_2 polymerization method in ideal slow addition mode. Moreover, Suzuki et al. [21] defined DB in the ring-opening multi-branching polymerization system as a unit ratio of the dendritic moiety to the total of the dendritic and linear ones (Eq. 4). The obtained DB_{suzuki} values were calculated to be between 0.40–0.50.

$$DB_{Suzuki} = \frac{D}{D + L} \tag{4}$$

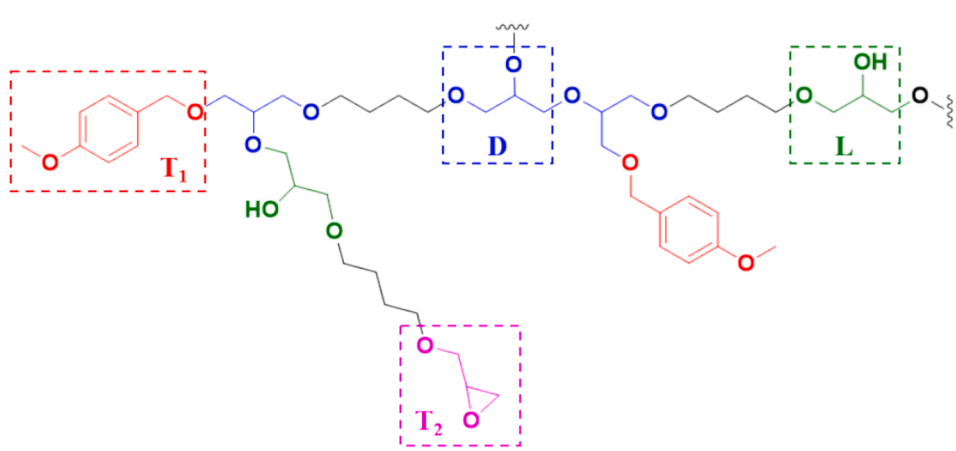
3.2. Synthesis of HBPEths with controlled main chain structures

According to the "$A_1 + B_2$"-ROP, HBPEths with controlled backbone could be straightforwardly prepared by employing biepoxy monomers featuring diverse linkages between both epoxy groups. Notably, only biepoxy monomers with high purity are suitable for the "$A_1 + B_2$"-ROP. Impurities in biepoxy monomers critically influence HBPEth

Table 2
Results of the synthesis of HBPEths with various main chain structures.[a]

Sample	[M]: [I3][b]	DB_{Frey}[c]	DB_{Suzuki}[c]	End group ratio [alkynyl]: [OH]	M_w (g/mol)[d]	$Đ_M$[d]
HBPEth-M2-I3	1.0: 1.5	0.66	0.49	49: 51	14,500	5.1
HBPEth-M3-I3	1.0: 1.1	–	–	54: 46	2900	1.4
HBPEth-M4-I3	1.0: 1.2	0.63	0.46	47: 53	8400	4.4
HBPEth-M5-I3	1.0: 1.2	0.58	0.45	48: 52	3300	2.6

[a] Conditions: polymerization was conducted at 40 °C for 24 h.
[b] Initiator was deprotonated by using NaH for 2.5 h in DMSO before adding M2.
[c] Calculated by ^{1}H NMR spectroscopy in DMSO-d_6.
[d] Determined by SEC in THF.



Scheme 3. Possible chemical structure of HBPEth-M1-I1.

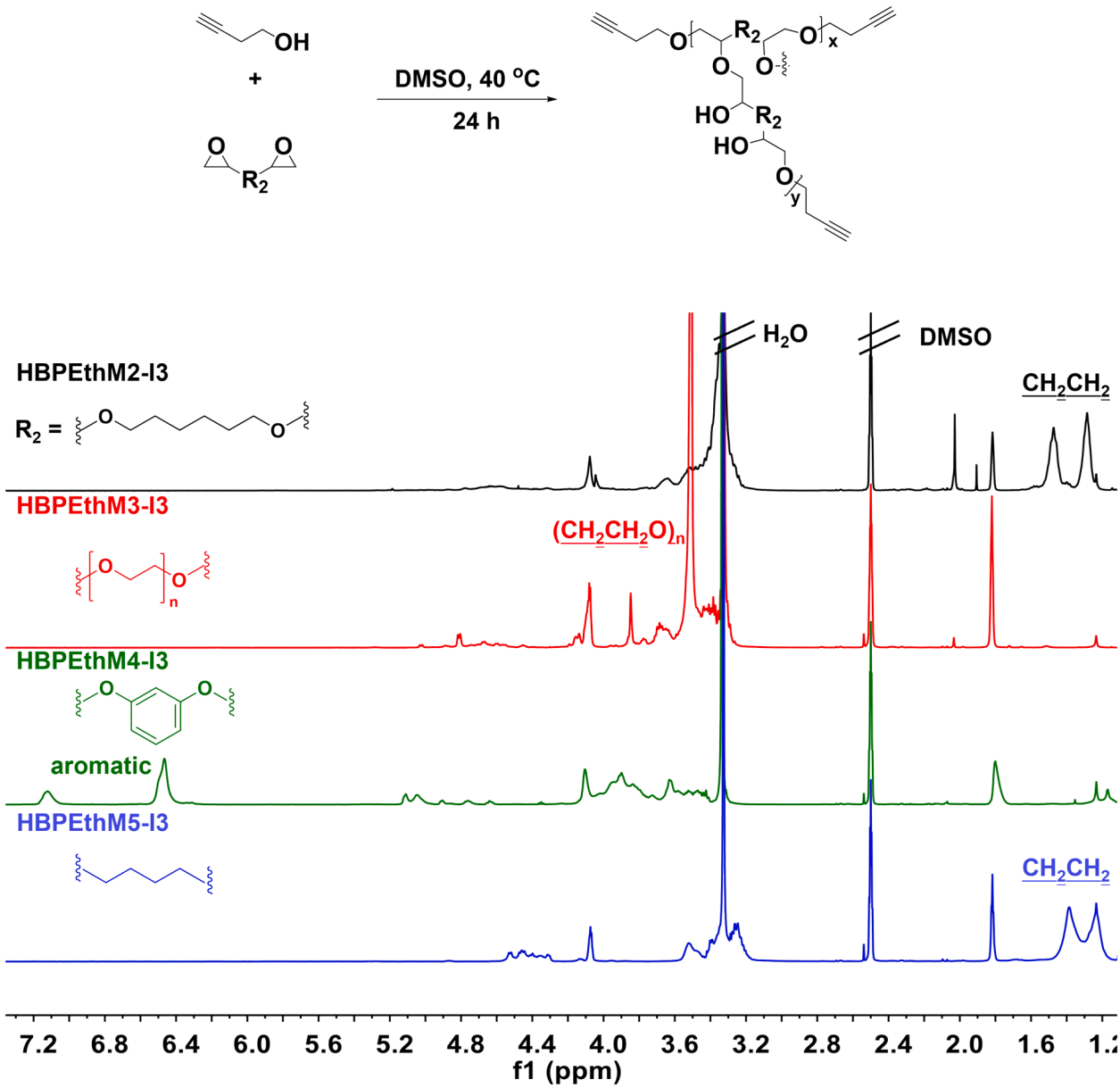


Fig. 4. ^{1}H NMR spectra of HBPEth with various main chain structures initiated by I3.

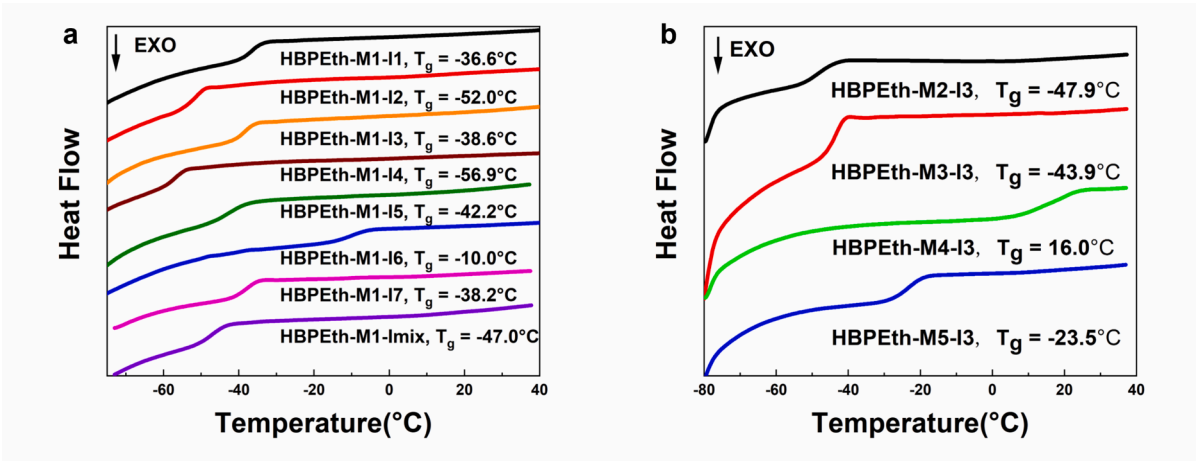


Fig. 5. DSC curves of HBPEths with different structures.

architecture and reaction kinetics. Monofunctional species (e.g. monoglycidyl ethers or diols) act as linear chain extenders or chain terminators during the anionic "A_1 + B_2"-ROP, limiting the formation of branching points and resulting in lower branching density. The "A_1 + B_2"-ROP was conducted at 40 °C in DMSO solution for 24 h using I3 as an initiator. Due to the different activities of biepoxy monomers, the initial feed ratio was varied from 1.0: 1.0 to 1.0: 1.5 to achieve high epoxide conversion while avoiding cross-linking side reactions. As illustrated in Table 2, the obtained HBPEths exhibited molar masses ranging from 2900 to 32000 g/mol. It is evident from Fig. 4 that the alkynyl end group can be clearly found at 1.80–1.90 ppm in ^{1}H NMR spectra, indicating successful initiation and subsequent polymerization of all biepoxy monomers (M1-M5) with different main chains. The absence of signals for unreacted epoxy groups at 2.6–3.0 ppm demonstrated a high epoxy conversion rate exceeding 99 % in all cases. Despite different initial feed ratios, the end group ratio of [alkynyl]: [OH] was approximately calculated to be 50: 50. Both the ^{1}H NMR (Fig. 4) and IR (Figs. S9-S12) spectra were consistent with the expected structures of HBPEth-M2-I3 – HBPEth-M5-I3. Similar to HBPEth-M1, DB values for HBPEths with different main chain structures were determined to be 0.58–0.66 and 0.45–0.49 based on the Frey and Suzuki methods, respectively. Due to the overlapping peaks of HBPEth-M3-I3 in the ^{1}H NMR spectrum, its DB value cannot be calculated. Moreover, the same preparation route was repeated by using 2,2,2-trifluoroethyl alcohol as initiator to validate the universality of the "A_1 + B_2"-ROP approach, and similar results were obtained (Table S2, Figs. S13-S18). These results demonstrate that the "A_1 + B_2"-ROP is the universal method for preparing HBPEths with controlled main chain and end group structures.

3.3. Thermal properties of HBPEths

The thermal properties of the resulting HBPEths were characterized using differential scanning calorimetry (DSC). As shown in Fig. 5, all HBPEths are amorphous and their glass transition temperatures (T_gs) are strongly influenced by their chemical structures. For HBPEth-M1s, the T_gs ranged from -57 °C to -10 °C depending on the rigidity of end groups (Fig. 5a). For HBPEth with identical backbone, a higher rigidity in end group structure led to a higher T_g for the corresponding HBPEth. Moreover, altering the HBPEth backbone by utilization of various biepoxy monomers had a significant impact on the thermal properties. As depicted in Fig. 5b, increasing the aliphatic chain length between both epoxy groups in B_2 monomers results in a decrease in T_g. The HBPEth-M4-I3 and HBPEth-M5-I3 containing rigid butylene or aromatic backbones displayed higher T_gs of -16.0 °C and -23.5 °C, respectively, in comparison to other HBPEths with relatively flexible chains such as HBPEth-M1-I3 and HBPEth-M3-I3 ranging from -38 °C to -48 °C. Compared to their linear or less branched polyether counterparts, the HBPEths exhibit significantly elevated T_gs. This enhancement can be attributed to their highly branched architecture and the abundance of hydroxyl groups, which facilitate extensive hydrogen bonding networks. A notable example is the contrast between linear poly(tetrahydrofuran) (PTHF) with a T_g of approximately -76 °C and HBPEth-M1s derivatives demonstrating substantially higher T_gs ranging from -10 °C to -57 °C. The observed elevation in T_g values across all synthesized HBPEths presents a distinct advantage for applications demanding enhanced thermal stability and dimensional integrity [40,41]. Furthermore, the "A_1 + B_2"-ROP strategy offers exceptional synthetic flexibility, enabling precise tuning of thermal properties through strategic selection of A_1 initiator and B_2 monomer units, thereby providing a versatile platform for developing materials with customized thermal characteristics tailored to specific application requirements.

4. Conclusion

In conclusion, a series of HBPEths with controllable end group and backbone structures were prepared by employing "A_1 + B_2"-ROP of biepoxy monomers under mild conditions. By using functional A_1 primary alcohol initiators, it is possible to directly introduce a single type or a mixture of functionalities, including fluoride, alkyne, alkene, azide, and furan groups, into the HBPEth structures, while by adjusting various B_2 monomers with different linkage between both epoxy groups the main chain structures could be well controlled. Under optimized polymerization conditions, HBPEths with molar masses ranging from 2600 to 30000 g/mol with expected DB values were obtained. Based on this strategy, a huge library of biepoxy monomers can be prepared by the reaction of beta-epoxypropylchloride and diols, and thus used in the "A_1 + B_2"-ROP approach for fabricating HBPEths with desired functionalities.

CRediT authorship contribution statement

Xinkai Zou: Writing – original draft, Investigation, Data curation. **Xiaofei Sun:** Investigation, Data curation. **Anxin Xu:** Investigation. **Yang Liu:** Investigation. **Wei Zhao:** Investigation. **Yingxia Zong:** Methodology. **Jingjiang Sun:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Qingfu Wang:** Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2025.113806.

Data availability

Data will be made available on request.

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